

DON'T PANIC

Safe Water Management, Explained

START HERE: THE CLOCK IS WATER

Food is a someday problem. Water is not. A person can go weeks without eating and about **three days** without drinking — less in heat, less if you're working, less if you're sick. When the taps go dry, the clock starts, and it is the fastest clock you own.

The good news: you are probably already storing water and don't know it. Your water heater is a tank of drinking water wearing a disguise. Your toilet tanks, your pipes, your freezer's ice — all water. And rain falls out of the sky for free.

This book assumes the water company is not coming. It covers what to stock before, where to find water after, and how to make found water safe to drink — with real gear when you have it, and with kitchen junk when you don't.

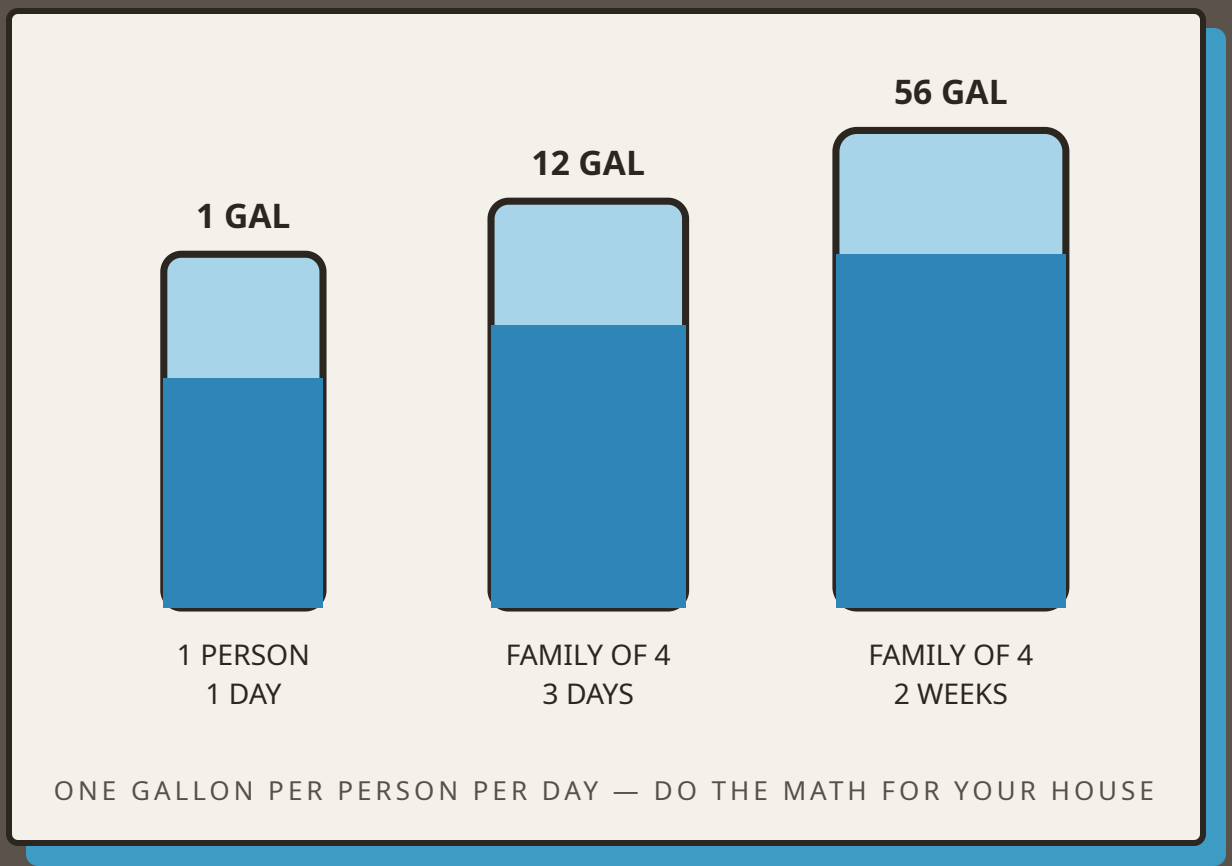
> **THE ONE RULE: clear is not clean.** Germs are invisible. Water that looks, smells, and tastes fine can still put you on the floor for a week. Every drop of found water gets treated before it gets drunk. No exceptions for "it looks fine."

1. HOW MUCH WATER A PERSON NEEDS

Plan on **one gallon per person per day**. About half of that is drinking; the rest covers cooking and the bare minimum of washing. In serious heat or hard work, drinking alone can take the whole gallon. Kids, nursing mothers, and sick people need more, not less.

Household	3 days	2 weeks
1 person	3 gal	14 gal
2 people	6 gal	28 gal
4 people	12 gal	56 gal

Three days is the floor, not the goal. Two weeks is a real target — it's one big barrel for a family of four. Pets drink too: roughly an ounce of water per pound of dog per day, so a 50-pound dog needs about a third of a gallon.



2. STOCK UP BEFORE YOU NEED IT

Stored water is the cheapest insurance in this whole series. Fill containers from the tap while the tap still works:

- **Five-to-seven-gallon jugs** with a spigot — the workhorse. Heavy when full (about 50 lbs), so fill them where they'll live.
- **A 55-gallon food-grade barrel** — the two-week answer for a family. Needs a siphon pump or hose to get water out.
- **Stackable water bricks** — smaller blocks that fit in closets and under beds.
- **A bathtub liner** — a food-grade bag that turns your tub into roughly 100 gallons of storage. You fill it when trouble is on the way, not before.
- **Cases of bottled water** — the grab-and-go layer. Fine as-is; rotate them like food.

How to store it so it's still good when you need it:

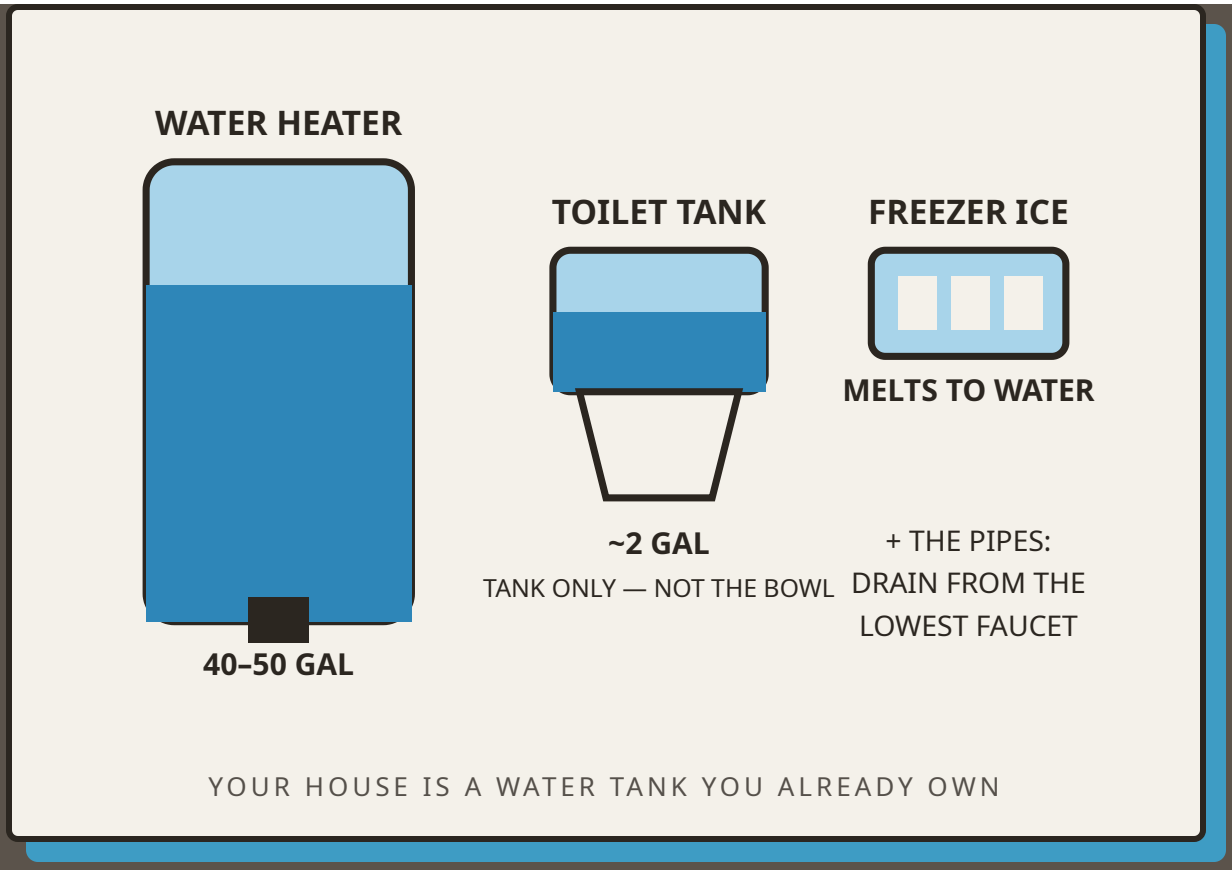
- Tap water that's already treated by a city needs nothing added. Just fill clean containers and cap them.
- Write the fill date on every container with tape and a marker. Use and refill stored water about every **6 to 12 months**.
- Store cool and dark, off bare concrete (set containers on wood), away from gasoline and chemicals — plastic breathes a little, and stored water slowly tastes like its neighbors.
- **Never** store drinking water in old milk jugs (the plastic breaks down and the seams split) or in anything that once held chemicals, ever.

> **Keep a water kit with the containers:** plain unscented bleach, an eyedropper, coffee filters, a squeeze filter, spare caps, a short food-grade hose or siphon pump, and a wrench that fits your water heater's drain. When the taps die is the wrong day to go shopping for a dropper.

3. THE WATER HIDING IN YOUR HOUSE

Before you go outside, drain the house. A typical home is quietly holding 50 or more gallons:

- **The water heater: 40–50 gallons.** This is the big one, and it's drinking water. Turn OFF its power (breaker) or gas first. Close the cold water inlet valve on top. Open a hot water tap somewhere upstairs so air can get in. Then open the drain spigot at the bottom and fill containers. The first cup may carry sediment — let it run until it clears.
- **Toilet tanks: about 2 gallons each.** The tank, never the bowl — tank water is the same water that was heading for your taps. Skip any tank with a blue cleaner puck in it.
- **The pipes themselves.** Open the highest faucet in the house to let air in, then drain from the lowest faucet into containers.
- **The freezer and pantry.** Ice melts into water. Canned fruit and vegetables swim in drinkable liquid.
- **Pool or hot tub water** is utility water — flushing toilets and washing. Treat it exactly like found water (Entry 5 onward) before even thinking about drinking it, and only as a last resort.


WATER HEATER
40–50 GAL
TOILET TANK
~2 GAL
TANK ONLY — NOT THE BOWL
FREEZER ICE
MELTS TO WATER
+ THE PIPES:
DRAIN FROM THE
LOWEST FAUCET
YOUR HOUSE IS A WATER TANK YOU ALREADY OWN

4. FOUND WATER OUTSIDE

When the house runs dry, water is still falling and flowing. In rough order of trust:

- **Rain** caught straight from the sky into a clean container — the cleanest found water there is. Roof runoff counts too but gets treated (Entry 11).
- **Moving water** — creeks and streams. Take it from where it flows, not from still edges, and walk upstream a bit from camps, farms, and roads if you can.
- **Lakes and ponds** — usable, but the stiller it is, the harder you treat it.
- **Dew and plant water** — real but tiny. A desperation measure, not a plan.

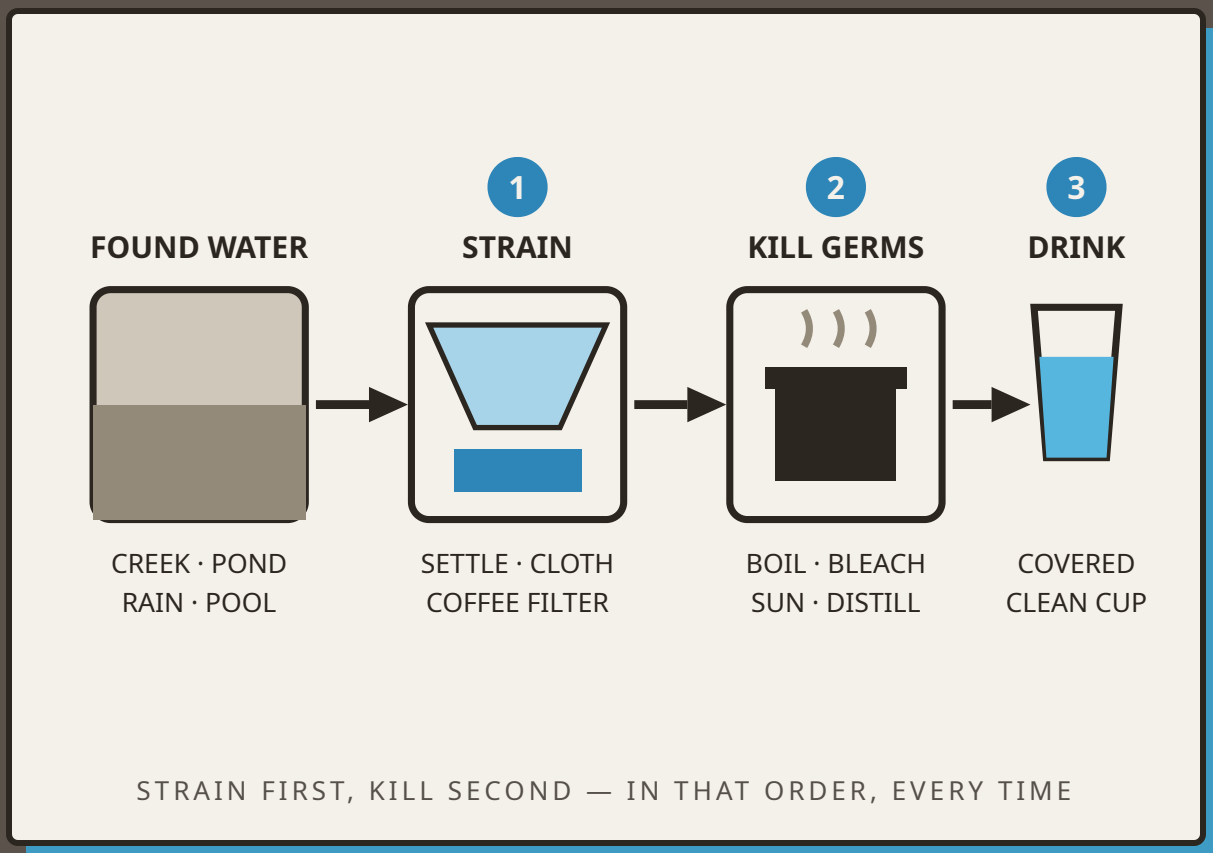
Whatever the source: take the clearest water you can find. Every treatment in this book works better on clear water, and a filter clogs slower. Scoop from below the surface, not the scum on top and not the mud on the bottom.

> **All found water is guilty until treated.** Animals drink from it, animals die in it, and everything uphill drains into it. The treatment order is always the same: strain out the chunks, then kill what's living in it. Next entry.

5. THE SAFE ORDER: STRAIN, THEN KILL

Every method in this book hangs on the same two steps, in this order:

1. **Strain it.** Let muddy water sit in a bucket until the dirt settles, pour the clearer top off, then run it through a coffee filter, a clean cloth, or a real filter. This takes out the chunks — and matters more than it looks, because disinfectant barely works in mud soup.
2. **Kill what's in it.** Boil it, bleach it, or bake it in the sun (Entries 7–10). Straining alone never makes water safe. Clear water can still be full of germs.
3. **Keep it covered.** Treated water goes in a clean, capped container and stays that way (Entry 12).



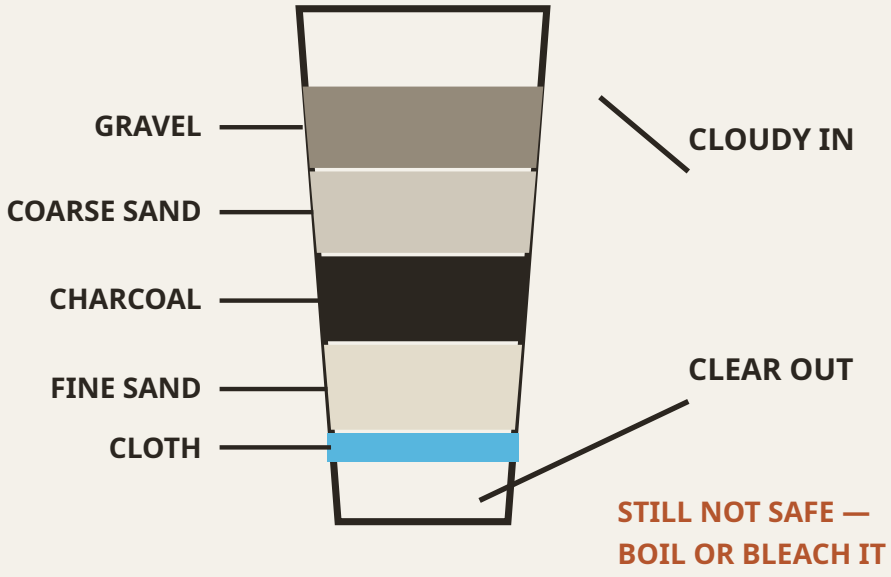
One thing straining and killing germs do *not* fix: chemicals, salt, and metals dissolved in the water. Those need distilling (Entry 9) — or a different water source (Entry 13).

6. A MAKESHIFT FILTER FROM JUNK

No filter in the house? Build the chunk-remover out of a bucket or a cut two-liter bottle. From top to bottom, the layers go:

1. **Gravel or pebbles** — catches the big stuff.
2. **Coarse sand.**
3. **Crushed charcoal** — plain wood charcoal from a fire, rinsed. (Not self-lighting briquettes — they're soaked in lighter fluid.) Charcoal also improves taste and grabs some chemicals.
4. **Fine sand.**
5. **A cloth at the bottom** to hold it all in.

Pour pond water in the top. What drips out the bottom should run clear. Pack the layers tight, let the first cup or two run through and throw it out, and rebuild the filter when it slows to a crawl.



LAYERS DO THE STRAINING — NEVER SKIP THE KILL STEP AFTER

Clear is not clean. This filter removes mud, bugs, and grit. It does not remove germs. Every drop that comes out of it still gets boiled, bleached, or sunned before drinking. A squeeze filter from a store (the kind hikers carry) is the same idea made better — it strains out bacteria and protozoa down to a size germs can't slip through, but most still let viruses and chemicals pass. Treat, then trust.

FIELD RECIPE · BUCKET FILTER

Makes: a strainer that turns muddy water clear. It does NOT make water safe — the kill step still follows.

Component	Count
Bucket, or a cut 2-liter bottle	×1
Gravel or pebbles	1 layer
Coarse sand	1 layer
Crushed wood charcoal — never self-lighting briquettes	1 layer
Fine sand	1 layer
Clean cloth	×1

1. Build from the bottom up: cloth across the bottom first, then fine sand, charcoal, coarse sand, and gravel on top.
2. Pour found water in the top. Throw out the first cup or two while it settles in.
3. What drips out is clear. Now boil it, bleach it, or sun it — clear is not clean.

7. BOILING: THE OLDEST KILL STEP

Heat is the disinfectant nobody can run out of, as long as there's fuel. The method is one sentence long:

> **Bring strained water to a full rolling boil — big churning bubbles, not a few lazy ones — and keep it there for 1 full minute.** Up in the mountains (above about 5,000 feet), make it 3 minutes, because water boils cooler up there.

- Keep the lid on while it cools. A lid keeps new germs out and fuel bills down.
- Boiled water tastes flat — the air got boiled out of it. Pour it back and forth between two clean containers a few times to put the air, and the taste, back.
- Never top up a container of treated water with untreated water, and never dip a used cup into the storage pot. One dirty dip undoes the whole boil.

Boiling kills germs dead — bacteria, viruses, protozoa, all of it. What it does *not* do is remove anything dissolved: salt stays, metals stay, chemicals stay, and as water boils away they actually get stronger. Dissolved problems are distilling's job, next entries down.

8. BLEACH: THE DROPPER METHOD

No fuel to spare? Plain household bleach disinfects without heat. It has to be the right bleach: **regular, unscented**, labeled sodium hypochlorite around 5–8%. No scents, no "splash-less," no color-safe. Those are laundry products, not disinfectants.

Water	Clear water	Cloudy water*
1 quart	2 drops	4 drops
1 gallon	8 drops	16 drops
5 gallons	½ teaspoon	1 teaspoon

*Cloudy water gets strained first (Entry 5), then double-dosed. Mud eats bleach.

1. Add the drops, stir, and wait **30 minutes**.
2. Sniff it. Treated water should smell faintly like a swimming pool. That smell is the proof there's bleach left over after killing everything.
3. No chlorine smell at all? Dose it again and wait 15 more minutes. Still nothing? The bleach may be dead — find a fresher bottle.

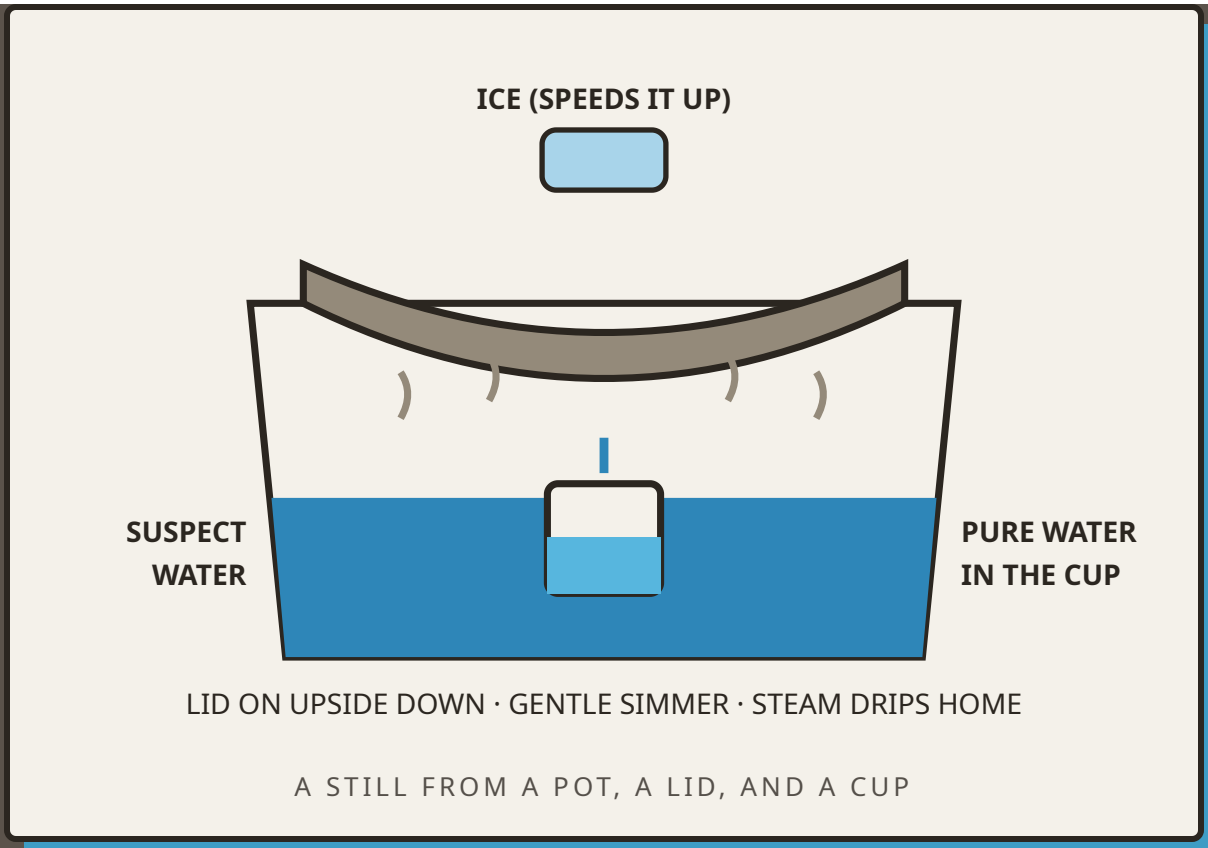
> **Bleach has a shelf life.** It starts weakening the day it's made and is noticeably tired after about 6 months. Write the purchase date on the bottle, rotate it like food, and keep a spare. The long-storage backup is **calcium hypochlorite pool shock** (65–70%, plain, no algaecides or fragrances) — the dry granules keep for years. To use it: stir **1 heaping teaspoon into 2 gallons of water** to make a stock solution, then add about **2½ tablespoons of stock per gallon** of water to treat, and wait the same 30 minutes. Label the stock bottle like it matters — because undiluted, this stuff is genuinely hazardous. Gloves on, keep it dry, keep it away from kids and from anything flammable.

9. DISTILLING ON A STOVE

Distilling is the heavy tool. It doesn't just kill germs — it leaves the dissolved stuff behind: salt, metals, and most chemicals stay in the pot while pure steam rises, cools, and drips into your cup. It's the only kitchen method that makes **seawater** drinkable, and the last resort for water you don't trust at all.

The setup is a pot, a lid, and a cup:

1. Put your suspect water in a big pot.
2. Set a clean cup or small bowl **inside** the pot, standing on an upside-down saucer, a canning ring, or a small rack so it sits steady and its rim stays above the waterline. The cup starts empty — it's the collector.
3. Put the lid on **upside down**, so its dome points into the pot and its middle is the lowest point, right over the cup.
4. Bring the water to a gentle simmer — not a violent boil. Steam rises, hits the cool lid, runs down the dome, and drips off the center into the cup.
5. Ice or a cold wet cloth on top of the lid speeds it up a lot. Swap it as it melts.



- **Know the cost:** distilling is slow — expect a cup or two an hour — and it eats fuel the whole time. Save it for water nothing else can fix.
- Distilled water tastes empty and has no minerals. Fine for weeks; if it becomes your long-term supply, eat normally and it'll even out.
- Steam burns are real burns. Keep the pot stable, keep kids back, and never leave a burner running unwatched.
- The same trick works off a camp stove, a rocket stove, or a careful fire — anywhere you can hold a simmer.
- **The pit version** (no stove at all): dig a pit in damp ground, set a cup in the middle, cover with clear plastic weighted at the edges, and put a small stone in the center so the sheet dips over the cup. Sun drives moisture out of the ground, it condenses on the plastic and drips in. Honest warning: a pit still makes sips, not gallons. It's a survival trick, not a water supply.

FIELD RECIPE · STOVE STILL

Makes: pure water out of almost any water — even seawater. Slow, and it eats fuel.

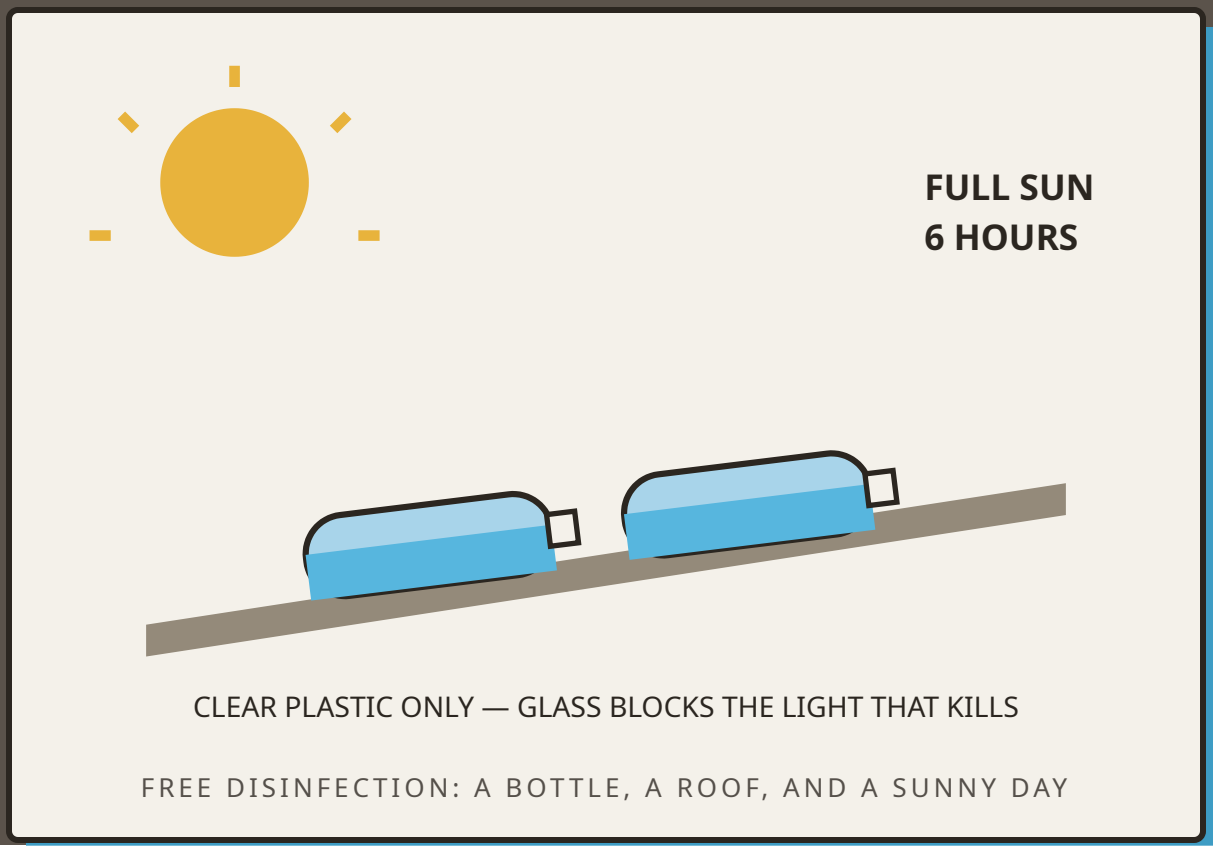
Component	Count
Big pot with a lid	×1
Clean cup or small bowl — the collector	×1
Upside-down saucer or canning ring — a stand for the cup	×1
Ice or a cold wet cloth (optional — speeds it up)	×1

1. Suspect water goes in the pot. The cup stands inside on its saucer, empty, rim above the waterline.
2. Lid on upside down, so its lowest point hangs over the cup.
3. Gentle simmer — never a hard boil. Steam condenses on the lid, runs to the middle, and drips into the cup.
4. Ice on the lid if you have it. Swap it as it melts.

10. SUN IN A BOTTLE

The sun is a disinfectant, and a clear plastic bottle is the machine. Relief workers call it SODIS — solar disinfection — and it works on the same germs boiling does, using nothing but light and time:

1. Use **clear plastic** bottles (the PET kind soda comes in), 2 liters or smaller, labels off. Glass blocks the ultraviolet that does the work — plastic only. Scratched-up bottles work less well.
2. Fill with strained, reasonably clear water. Cloudy water shades its own germs.
3. Shake the bottle hard to mix air in — the oxygen helps the sunlight kill — then top it off.
4. Lay bottles on their sides in full sun. A dark roof or a sheet of metal underneath gets them hot as well as lit, and heat plus light kills faster.
5. **6 hours of strong sun.** If the sky is more than half cloudy, give them 2 full days.
6. Drink straight from the bottle, or pour without touching the threads — hands on the bottle mouth re-dirty the water you just cleaned.



FIELD RECIPE · SUN BOTTLES (SODIS)

Makes: disinfected water with no fuel at all — just sun and time.

Component	Count
Clear plastic (PET) bottles, 2 liters or smaller	×4
A dark roof or a sheet of metal	×1

1. Fill with strained, clear-ish water. Shake hard to mix air in, then top the bottle off.
2. Lay the bottles on their sides in full sun, on the dark surface.
3. Six hours of strong sun — or two full days if the sky is more than half cloudy.
4. Drink from the bottle, or pour without touching the threads.

11. CATCHING RAIN ON PURPOSE

Your roof is a giant funnel you already own. A rain barrel under a downspout turns a storm into stored water:

- **Screen the top.** A fine mesh keeps out leaves, bugs, and mosquitoes — and keeps the barrel from becoming a mosquito farm.
- **Skip the first rain.** The first minutes of a storm wash dust, pollen, and bird droppings off the roof. If you can, let that first flush run off (a simple diverter pipe that fills first and then passes clean water on does this by itself), and catch what falls after.
- **Keep it dark and covered.** Light grows algae. A lid and a dark barrel keep the water boring, which is what you want.
- **Point the overflow away** from the house's foundation.
- Roof water is found water. It gets the full strain-and-kill treatment before drinking, every time — the roof is not a clean place.

The math is cheering: one inch of rain on 1,000 square feet of roof is roughly **600 gallons**. Even a small roof in a modest storm fills a barrel without trying.

FIELD RECIPE · RAIN BARREL WITH FIRST FLUSH

Makes: a roof-fed water store that skips the dirty first rinse.

Component	Count
Food-grade barrel with a lid	×1
Fine screen mesh	×1
Short pipe section — the first-flush diverter	×1
Spigot (optional)	×1

1. Barrel under the downspout, screen across the top to keep out leaves, bugs, and mosquitoes.

2. The diverter pipe catches the first minutes of rain — the roof wash — and only then passes water on to the barrel.
3. Overflow points away from the house. Lid stays on; the barrel stays dark.
4. Rain water is found water: strain it and kill what is in it before drinking.

12. KEEPING SAFE WATER SAFE

More treated water gets ruined in storage than was ever ruined at the source. The rules are boring and absolute:

- **Pour or pump — never dip.** A dipped cup carries yesterday's hands into today's water. A spigot or a pour keeps the inside of the container a clean zone.
- **Lids stay on.** Every container, all the time. Dust, bugs, and curious toddlers are all contaminants.
- **Label everything:** what's in it, when it was treated, and how. Tape and a marker. In the dark, in a hurry, memory lies and labels don't.
- **Two families of containers, never mixed:** DRINK water and WASH water, marked in big letters. The creek bucket never becomes the drinking jug.
- **Wash containers before filling:** soap and water, then a sanitizing rinse — about 1 teaspoon of bleach per quart of water — swished around, then air-dry upside down.

13. STOP SIGNS

Some water you walk away from, no matter how thirsty the day feels:

- **A rainbow sheen on the surface, or any smell of fuel, solvent, or chemicals.** That's petroleum or industrial runoff. Don't drink it — and know that even distilling can't be fully trusted here, because some fuel chemicals boil and travel right along with the steam. Find different water.
- **Bright green, scummy, paint-looking water.** Algae blooms can carry real poisons that boiling does not remove. Walk upstream or find another source.
- **Floodwater.** Assume it carries sewage and every chemical from every garage it passed through. Last-resort-only water, and even then: strain, distill if you possibly can, and know it's a gamble.
- **Seawater, straight.** It dehydrates you faster than drinking nothing — your kidneys spend more water removing the salt than the water brought in. Distill it or leave it.
- **A dead animal in or right at the source.** Move well upstream, or find other water.
- **Water that already made someone sick.** Stop. Re-treat it a stronger way, or change sources. Don't run the experiment twice.

Knowing which water to refuse is most of having the skill. Thirst argues loud and fast; the stop signs argue once, quietly, and they're right.

14. THE WHOLE BOOK ON ONE CARD

- One gallon per person per day. Two weeks stored is the goal.
- Drain the house first: heater, toilet tanks, pipes, ice.
- All found water is guilty until treated. Clear is not clean.
- The order never changes: strain the chunks, then kill the germs.
- Boil 1 minute rolling (3 in the mountains). Or bleach: 8 drops a gallon, wait 30, smell for chlorine.
- Makeshift filter = gravel, sand, charcoal, sand, cloth. It clears water; it does not clean it.
- Distilling fixes what nothing else does — salt, metals, seawater. Slow and fuel-hungry.
- Sun in a clear plastic bottle, 6 hours, disinfects for free.
- Store treated water covered, labeled, and never dipped into.
- Sheen, fuel smell, green scum, floodwater, straight seawater: stop signs. Walk away.

ALL BOOKS, A–Z